USE: courseManFinalV1.use

File Edit State View Plugins Help

Model Validator
View
CAP
Transform Model into CAP Model

CourseMan
- Data types
- Classes
  - Student
  - CourseModule
  - CourseOffering
  - Program
  - AcademicTerm
  - TermRecord
  - Instructor
  - Enrolment
  - ProgramEnrolmer
  - HelpRequest
- Associations
  - Offers
  - Prerequisites
  - Corequisites
  - RequiredPreviou
  - Exclusions
  - CourseProgram
  - StudentTerm
  - TermOffering
  - TermRecordTern
  - Teaching
  - RequestStudent
- Invariants
- Pre-/Postconditions
  - pre Date::pre1
  - pre Date::pre2
  - pre Time::pre3
  - pre Time::pre4
  - pre Time::pre5
- Query Operations
  - AcademicTerm::
  - CourseOffering::
  - Enrolment::grade
  - Enrolment::isPas
  - Enrolment::modu
  - Enrolment::term
  - ProgramEnrolmer
  - Student::complet
  - Student::complet
  - Student::enrolme
  - Student::enrolme

1

Create CAP
Create a new CAP specification

Manage CAP
View, edit, delete and open diagrams

Convert Annotation to OCL
Convert from Annotation to OCL

Parse CAP from File
Parse CAP specification from file

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CAP Annotation to OCL Converter

OCL Metadata
Context: Student
Name: Invariant

Annotation Input
```
    @AttrCond(attr =
offering.semester,
matchAttr =
currentSemester)
  },
  sumAttr = credits,
  fixAttr =
self.totalCredits
)
```

Result
SumConstraint: TYPE1

Generated OCL
```
context Student
inv Invariant:
  let s : Integer =
      self.enrolments
          ->select(e |
e.offering.semester =
self.currentSemester)
          ->collect(e |
e.course.credits)
          ->sum()
  in
```

Convert →

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Create CAP

CAP Name: SumConstraint

CAP Grammar (.use)
```
inv cm01_CoreCreditsRequirement:
  self.status = StudentStatus::GRADUATED implies
    self.enrolment ->select(e | e.grade >='C'
                        and e.offering.module.type = CourseType::CORE
        ->collect(e | e.offering.module.credits)->sum()
        >= self.programEnrolment.program.minCoreCredits
```

Description

Load .use file | Add CAP

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CAP File Parser

Upload File

Input File
```
59 @SumConstraint(
60          assocCls =
61          rolePath =
62          collect = {
63            @AttrCond
64          },
65          sumAttr = c
66          maxAttr = m
```

Output File
```
36 constraints
37
38 context ContextClass
39 inv ConstraintName:
40   self.status = Stud
41 let v : Integer =
42       self.enrolment
43           ->select(e
```

Validate .use | Download File

Convert →